

Partial_02 Report

The program that I developed for my second partial in the class of Multimedia and Computational Graphics was a code in Java that created a video that mixes AI and real life photos. The program receives a set of images that are chosen by the user and then processes them to generate a final video. The program reads the metadata of the images such as the date and the GPS location and according to this it organizes them chronologically. Then the program creates a video and prioritizes portrait format, the video has an introductory image, the submitted images with specific audios made by AI and the final scene is a map comparing the location of the first and last photo and an inspirational phrase.

The program was designed using Object-Oriented Programming principles. It was divided into multiple classes, each one responsible for a specific task. The Main class is the one that initializes the execution of the program. The ProjectRunner class helps to assemble the final video. The MediaItem class stores the metadata such as file path, date, and GPS coordinates, MetadataUtils extracts this metadata. The AIService class is the one that communicates with the Gemini API to generate text descriptions and phrases. Finally, the VideoCreator class combines audio and video into a final video using FFmpeg.

The process begins by reading all files from the input_media folder. The program uses metadata extraction tools to obtain information from the photos like the date and its geographic coordinates. This information is stored inside MediaItem objects and they are sorted from oldest to newest. Then the program creates a short AI description for each photo using the Gemini API. This description is then converted into audio. Then FFmpeg creates individual video clips for each item which will then be joined.

A very interesting part of the program is the generation of the final map. The program uses the GPS coordinates from the first and last media items and calculates their positions on a world map image. It then draws dots to indicate the beginning and ending locations of the journey and joins them. This converts coordinates to pixel positions on the map image. The map also includes an inspirational phrase generated by the AI on the bottom.

The program then combines all these elements into a final video. This project is a demonstration of multimedia processing, API integration, and object-oriented design. It combines different technologies such as Java, external command execution, metadata extraction, and video processing tools. The program meets the basic requirements of the assignment by generating a complete video from a set of selected files. One thing I would definitely like to improve about this project would be to make the AI introductory image better and get out errors using the API as soon as possible because they cost me a lot of time.